Google Developer Group
Bandung Institute of Technology

Team name	Tide Ledger
Project title	RapiLPJ: Pre-Submission Audit for Student Organization Activity Reports
Selected case	Case 1: Smarter Campus and Learning
Members	Vincentius Bryan Kwandou (leader), Valentino Filemon Limer, Putra Purwanugraha Tengbunan, Marvel Harjosetio

Abstract

1. Problem. Untrained student treasurers manage campus funds under strict bylaws: physical stamps, balanced ledgers, meal-attendance ratios, revenue stamps over Rp5,000,000 (Law 10/2020), taxes, and deadlines. Incomplete submissions freeze campus funding and peer learning [6], trapping study hours in mechanical revisions without fiscal judgment.

2. Intended Users. Primary: club and BEM treasurers and secretaries (maintaining attendance and verifying figures). Secondary: faculty verifiers (Biro Kemahasiswaan, student auditors) conducting desk reviews to log audit trails without releasing funds.

3. Supporting Observations or Evidence. Siada and Tungadi (2021) document unstandardized activity documentation at BKAM UAJM [1]. Guidelines diverge: UB requires physical stamps and balanced ledgers [2]; UGM enforces Rp10,000 stamps above Rp5,000,000 under Law 10/2020 [3]; ITS mandates 10-day deadlines [4] and taxes [5]; FILKOM UB notes recurring withholding failures [6].

4. Proposed Solution and Core User Flow. RapiLPJ turns rejection into a gated practice run. (1) Perceive, then confirm. Gemini 2.0 Flash reads receipt photos into JSON (date, vendor, nominal, line type) without arithmetic. Fields sit beside the photo for treasurer confirmation; unconfirmed JSON cannot enter the ledger, preventing misread Rp50,000 from auditing as Rp500,000. (2) Deterministic kernel on confirmed fields in portable .lpj: duplicate invoices, dates outside event windows, missing stamps over Rp5,000,000, and variances above 10%. Meal portions and PPh 21 are not inferred: two faculty advisors sign a UAJM rule card. Meals fail when portions exceed attendance multiplied by card allowances; honoraria fail when withholding rates differ from card rates. Missing card rates return 'preset gap'; the engine never invents rates. (3) Gated literacy. Infractions lock export until micro-quizzes match statutory keys (Rp5,000,000 exactly is not dutiable; nominals above are dutiable under Law 10/2020, Art. 3). Templates draft variance addenda without LLM involvement. (4) Desk-check file, not disbursement. Export yields indexed PDF dossiers plus SHA-256 hashes over confirmed ledgers, quizzes, and addenda. Verifiers recompute hashes locally, recording 'lolos desk-check' or notes into .lpj files. That returned file is the record; hash matching is instant, notes reflect reading time, and no control releases funds.

5. How It Differs from Existing Alternatives. Pure LLMs miscalculate ledgers; spreadsheets lack hash-binding, statutory checks, and quiz gates; portals merely collect files. RapiLPJ enforces a strict boundary: Gemini perceives, humans confirm figures, while arithmetic, statutory checks, quiz keys, and addenda remain 100% deterministic client-side code.

6. Planned Technologies. Perception: Gemini 2.0 Flash Multimodal (client BYOK, JSON-only, single-session). Kernel: open-source JavaScript, locked templates, FileReader, .lpj JSON, SHA-256. Presentation: accessible bilingual HTML5/CSS3 static on Vercel without databases; receipts route directly to Gemini under user key without app archival.

7. Expected Benefit and Development Plan. Evaluating 30 treasurers and 2 UAJM advisors across five metrics: treasurer minutes, mechanical defects before/after kernel, fields corrected after Gemini, quiz scores before/after, and verifier minutes. Semifinal benchmarks single-receipt perception, confirmation, signed card, quiz gate, and desk-checks; Final Day delivers multi-page batching and QR re-checks.

Source Acknowledgments. [1] P. Siada & A. L. Tungadi, TEMATIKA J. Penelit. Tek. Inform. Sist. Inf. UAJM, 2021, doi:10.56963/tematika.v9i1.119 (studi sistem informasi portofolio kemahasiswaan BKAM UAJM). [2] Pedoman LPJ, FIB Universitas Brawijaya, fib.ub.ac.id. [3] Prosedur Proposal & LPJ SIPRO, FEB UGM & UU Bea Meterai No. 10/2020. [4] SOP HIMA, Desain Interior ITS, its.ac.id. [5] SOP LPJ Mahasiswa Biologi, ITS, its.ac.id. [6] Surat Edaran Mekanisme LPJ Ormawa, FILKOM UB, 2026. AI disclosure: Claude and Gemini supported research, deterministic engine code, and drafting.